# Splitting Water with Electricity

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| ■ | **Learning Objectives**<br>Understand what electrolysis is, how it works, and where it is used in real life. |
| ■<br>■ | **Duration**<br>45–60 minutes (can be split across two periods) |
| ■ | **Subject Links**<br>Science, Chemistry, Environmental Studies |
| ■ | **Hands-On?**<br>Yes – safe experiment with a 9V battery, pencils & salt water |

■ **Big Idea:** Water ($H_2O$) can be split into two gases — hydrogen ($H_2$) and oxygen ($O_2$) — by passing electricity through it. This process is called **electrolysis**, and it has exciting real-world uses from clean energy to making jewellery shine!

## ■ Section 1 – What Is Water Made Of?

### Prior Knowledge Activation

Before the lesson begins, ask students:

■ **What is water?** Where do we find it?
■ **What is electricity?** Name three things that use electricity at home.
■ **Have you ever seen bubbles form** in water? What causes them?

### Water's Secret Recipe

Everything around us — air, soil, food, even our own bodies — is made of tiny building blocks called **atoms**. Atoms join together to form **molecules**. A water molecule is one of the simplest and most important molecules on Earth.

| Atom | Symbol | Count in H■O | Fun Fact |
|---|---|---|---|
| Hydrogen | **H** | **2** | Lightest element in the universe! |
| Oxygen | **O** | **1** | We breathe it to stay alive. |

*Table 1 – Atoms that make up one water molecule ($H_2O$)*

*Think of it this way: making a water molecule is like making roti dough — you need two parts atta (hydrogen) and one part water (oxygen) to create something new!*

**■ Key Term: Molecule**

A molecule is the smallest unit of a substance that keeps all the properties of that substance. A water molecule is made of 2 hydrogen atoms and 1 oxygen atom, written as $H_2O$.

## ■ Section 2 – What Is Electrolysis?

Electrolysis sounds like a big word, but it simply means **using electricity to break apart a substance**. In our case, we use electricity to break water ($H_2O$) into its two parts: hydrogen gas ($H_2$) and oxygen gas ($O_2$).

■ **Analogy — Scissors & Rakhis:**
Imagine the bonds between hydrogen and oxygen atoms in a water molecule are like colourful rakhis (threads) tying them together. Electricity acts like a pair of scissors that snips those threads, setting the hydrogen and oxygen free as separate gases!

### The Chemical Equation

$$2H_2O \rightarrow 2H_2 + O_2$$

**2 water molecules → 2 hydrogen gas molecules + 1 oxygen gas molecule**

### Breaking It Down Step by Step

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| **Step 1** | Electricity enters the water through conductors called **electrodes**. |
| **Step 2** | The electric current provides energy to break the bonds in $H_2O$ molecules. |
| **Step 3** | At the negative electrode (**cathode**), hydrogen gas bubbles form. |
| **Step 4** | At the positive electrode (**anode**), oxygen gas bubbles form. |
| **Step 5** | The gases rise to the surface as bubbles — we can see and collect them! |

■ **Key Term: Electrolysis**

The process of using electrical energy to drive a chemical reaction that would not happen on its own. The word comes from Greek: *electro* (electricity) + *lysis* (to loosen or split).

## ■ Section 3 – Components of an Electrolysis Setup

### ■ Battery (9V)

The **power source**. It pushes electric current through the circuit. We use a 9V battery because it is safe for classroom experiments. The battery has a positive (+) terminal and a negative (–) terminal.

■ *Teacher Tip: Do NOT use batteries above 9V — higher voltage can be dangerous.*

### ═■ Pencils (Electrodes)

The two pencils act as **electrodes** — they are the entry and exit points for electricity into the water. The carbon (graphite) inside pencils conducts electricity well and does not rust, making them ideal for this experiment.

■ *Teacher Tip: Connect the pencil connected to the (+) terminal → anode. The (–) pencil → cathode.*

### ■ Water

Pure water is actually a poor conductor of electricity. We need to add a substance to help current flow. The water provides the $H_2O$ molecules that will be split.

■ *Teacher Tip: Use tap water rather than distilled water — tap water already has dissolved minerals that help.*

### ■ Salt (Electrolyte)

Salt (sodium chloride, NaCl) dissolves in water and breaks into ions (charged particles) that carry the electric current. Without an electrolyte, very little current flows and almost no bubbles form. Think of it like adding a conductor highway for the electricity!

■ *Teacher Tip: Baking soda also works as an electrolyte and is safe for classroom use.*

| Electrode | Electrolyte | Cathode | Anode |
|---|---|---|---|
| A conductor through which electricity enters or leaves a substance. | A substance that conducts electricity when dissolved in water (e.g. salt). | The negative (–) electrode. Hydrogen forms here. | The positive (+) electrode. Oxygen forms here. |

*Table 2 – Key Vocabulary for Electrolysis*

## ■ Section 4 – Hands-On Experiment

■■ **Safety Rules — Read Before Starting!**

- Always have an **adult present** during the experiment.
- Use ONLY a **9V battery**. Never use a mains plug or higher-voltage source.
- Do **not touch bare wires** while the circuit is connected.
- Do **not taste** the water.
- Keep all materials **away from faces and eyes**.

### Materials Needed

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| ✔ 1 clear glass or plastic cup (300 mL) |
| ✔ Tap water (about 250 mL) |
| ✔ 2 teaspoons of table salt or baking soda |
| ✔ 2 pencils (sharpened at both ends to expose graphite) |
| ✔ 1 nine-volt (9V) battery |
| ✔ 2 pieces of insulated copper wire (15 cm each, ends stripped) |
| ✔ Electrical tape (optional, for securing connections) |
| ✔ A spoon for stirring |

### Step-by-Step Procedure

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| **1** | Fill the glass with approximately 250 mL of tap water. |
| **2** | Add 2 teaspoons of salt (or baking soda) and stir until fully dissolved. This creates your electrolyte solution. |
| **3** | Strip the ends of both copper wires using a knife (adult only) or buy pre-stripped wires. |
| **4** | Wrap one wire end around the graphite tip of each pencil and secure with tape. |
| **5** | Connect the free end of one wire to the (+) terminal of the 9V battery, and the other to the (–) terminal. |
| **6** | Carefully lower both pencil tips into the salt water, keeping the pencils about 2 cm apart. Do not let the wires touch the water. |

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| 7 | Observe! Bubbles should begin forming within seconds. Count and compare bubbles on each pencil. |
| 8 | Record your observations in the table below. |

## ■ Section 5 – Observations & Analysis

### Observation Table

| Observation | Negative Electrode (–) Cathode | Positive Electrode (+) Anode |
|---|---|---|
| Number of bubbles | ________ (more) | ________ (fewer) |
| Gas produced | Hydrogen (H■) | Oxygen (O■) |
| Colour of gas | Colourless | Colourless |
| Bubble size | Small & rapid | Larger, slower |
| Your observations… | ______________________________ | ______________________________ |

*Table 3 – Experiment Observations (students fill in highlighted rows)*

### Why More Bubbles at the Negative Electrode?

For every 1 molecule of oxygen ($O_2$) produced at the positive electrode, **2 molecules of hydrogen ($H_2$)** are produced at the negative electrode. Since hydrogen is produced in a 2:1 ratio, you will always see roughly twice as many bubbles at the cathode. This is a direct result of the balanced chemical equation!

$$2H_2O \rightarrow 2H_2 + O_2$$

**2 parts hydrogen : 1 part oxygen → double the bubbles at the cathode!**

### Discussion Questions (in pairs or groups)

1. What do you think would happen if you used **more salt**? Would there be more bubbles?
2. What if you used a **stronger battery** (e.g. 12V)? Would the process be faster?
3. Why do you think we cannot use pure, distilled water for this experiment?
4. Could we reverse the process — put hydrogen and oxygen together to make water? (Hint: This is how hydrogen fuel cells work!)

**■ Key Term: Cathode & Anode**

**Cathode** = negative electrode — electrons flow *into* the solution here. Hydrogen gas ($H_2$) is produced.
**Anode** = positive electrode — electrons leave the solution here. Oxygen gas ($O_2$) is produced.

## ■ Section 6 – Real-World Uses of Electrolysis in India ■■

Electrolysis is not just a classroom experiment — it is used in industries, hospitals, and even defence applications all across India. Let us look at some exciting examples!

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**Green Hydrogen Buses**

■

**Medical Oxygen Cylinders**

■

**Gold & Silver Electroplating**

■

**INS Arihant Submarine**

■ **Did You Know?** India has set a target to produce 5 million tonnes of green hydrogen per year by 2030 under the **National Green Hydrogen Mission**. Electrolysis will be the key technology to achieve this goal — and students learning this today could be the engineers who make it happen!

## ■ Section 7 – Quick Quiz

### Q1. What does electrolysis split?

a) Milk
b) Water ✓
c) Salt
d) Electricity

■ **Answer:** b) Water — electrolysis uses electricity to split H■O into H■ and O■.

### Q2. What gas forms at the NEGATIVE electrode (cathode)?

a) Oxygen
b) Carbon dioxide
c) Hydrogen ✓
d) Nitrogen

■ **Answer:** c) Hydrogen (H■) — electrons reduce water to form hydrogen at the cathode.

### Q3. Why do we add salt to the water?

a) For taste
b) To colour it
c) To conduct electricity ✓
d) To make it heavier

■ **Answer:** c) Salt dissolves into ions that carry the electric current through the water.

### Q4. What are the conductors that electricity enters and leaves through called?

a) Electrolytes
b) Electrodes ✓
c) Electrons
d) Elements

■ **Answer:** b) Electrodes — in our experiment, the pencil tips are the electrodes.

### Q5. At which electrode do you see MORE bubbles, and why?

a) Positive — more oxygen forms
b) Negative — twice as much hydrogen forms ✓
c) Both have the same number of bubbles
d) The one closer to the battery

■ **Answer:** b) Negative (cathode) — hydrogen is produced at twice the rate of oxygen.

## ■ Section 8 – Homework Challenge ■

**Your Homework Mission:**

1. Ask a parent or elder at home: *'Have you seen gold plating or silver plating done? Where?'*

2. Find out if any jewellers in your city or town use electroplating. If possible, visit a jewellery shop and ask!

3. **Draw what you imagine the electroplating setup looks like** — the tank, the jewellery, the electricity source. Be creative!

4. Write 2–3 sentences about what you learned and bring your drawing to share with the class.

✏■ *My Electroplating Drawing*

## ■ Lesson Summary

- ■ **Water (H■O)** — is made of 2 hydrogen atoms and 1 oxygen atom bonded together.
- ■ **Electrolysis** — uses electricity to break those bonds and separate the gases.
- ✏■ **Electrodes** — are the pencil conductors through which electricity enters the water.
- ■ **Electrolyte** — (salt) helps electricity flow through the water.
- ■ **Cathode (–)** — produces hydrogen gas (H■) — more bubbles!
- ■ **Anode (+)** — produces oxygen gas (O■) — fewer bubbles.
- ■ **Real Uses** — green hydrogen fuel, hospital oxygen, jewellery plating, submarines.

■■ **Science is not just in textbooks — it is in the buses we ride, the jewellery we wear, and the oxygen we breathe. Keep exploring!**